Question: Define the term "society" and discuss its fundamental elements with relevant examples. (5 Marks)

Answer:

The term "society" originates from the Latin word "socious," which translates to companionship or friendship. Sociologically, it refers to a group of people who inhabit a defined geographic territory and share a distinct culture. Anthony Giddens refines this by defining society as a group of individuals subject to a common system of political authority who are conscious of possessing a distinct identity from other groups. Morris Ginsberg similarly defines it as a collection of individuals united by certain relations or modes of behavior that mark them off from others who do not enter into those relations.

For a society to exist and function, it must possess four fundamental elements:

- Likeness: This is the absolute cornerstone of social identity. Without a sense of being "alike," there can be no recognition of a shared identity or community. This likeness allows for mutual understanding, social cohesion, and cooperation. Examples include a shared language, common customs, values, and core beliefs.
- Difference: While likeness provides the foundation and unites people, internal differences drive social interaction and reciprocity. If everyone were exactly the same, there would be no specialization, exchange, or complex relationships. The division of labor—where hunters, farmers, and artisans specialize in different tasks and rely on each other's distinct skills—is a prime example.
- Inter-dependence: Society cannot survive without cooperation. Individuals must rely on one another to meet their daily survival and well-being needs, establishing a complex system of mutual obligations and support. For example, a modern city dweller depends entirely on farmers for food, engineers for utility infrastructure, and doctors for healthcare.
- Multiplicity of relationships: A society is not defined by a single social bond but by a vast, dynamic web of thousands of overlapping relationships. These include familial, economic, political, religious, friendly, and professional connections that make social life rich and dynamic.

Question: Explain the unilinear theory of social evolution proposed by E.B. Tylor and Lewis Henry Morgan, outlining the characteristics of Savagery and Barbarism. (5 Marks)

Answer:

The 19th-century theory of social evolution posited that human societies develop gradually, transitioning over time from simple, homogeneous forms into complex, highly differentiated structures. Pioneered by Sir Edward Burnett Tylor and Lewis Henry Morgan, the core tenet of this classical evolutionist school is that social evolution is unilinear—meaning all human societies must progress through the exact same developmental stages in the exact same sequence.

Morgan and Tylor divided human progress into three major sequential stages:

- Savagery: This stage represents the earliest phases of human existence. It is further subdivided into Lower Savagery (before the discovery of fire, where tree-dwelling humans subsisted on simple gathered wild fruits and nuts, developing articulate speech), Middle Savagery (initiated by the mastery of fire and fish consumption, which allowed humans to migrate across diverse climates and became less dependent on specific locations, though marked by periods of cannibalism due to food uncertainty), and Upper Savagery (characterized by the invention of the bow and arrow, which revolutionized hunting and allowed early settled village life with wooden vessels and tools).
- Barbarism: This stage marks the transition to food production and is subdivided into Lower Barbarism (commenced with plant domestication—the Neolithic Revolution—and the invention of pottery to store and cook food, along with matrilineal clan systems), Middle Barbarism (defined by animal domestication providing reliable milk and meat, horticultures in some regions, a transition toward patrilineal descent, and individual property ownership), and Upper Barbarism (marked by the smelting of metals, leading humanity through the Bronze and Iron Ages with superior tools and weapons).
- Civilization: The final, current stage of human progress, characterized by intensive agriculture (producing surplus food), industrialization, capitalism, clear social stratification, and the emergence of organized political states with laws and a monopoly on force.

Question: Discuss the long-term transformation of human society, comparing the key characteristics of Primitive, Industrial, and Post-Industrial societies. (5 Marks)

Answer:

Social transformation refers to the long-term, structural changes in how societies organize their labor, technology, and social institutions. Driven by technological innovations, population pressures, and economic expansion, societies have continuously modernized, transforming from simple subsistence systems to highly complex technological networks.

The structural characteristics of the three landmark societal stages are compared as follows:

- Primitive (Hunting & Gathering) Societies: Existing from roughly 200,000 years ago to 10,000 BCE, these nomadic bands of 20 to 50 people depended entirely on wild plants and animals. They operated under a subsistence economy with simple stone, wood, and bone tools. Because there was no surplus wealth, these societies were highly egalitarian, featuring minimal social inequality, kinship-based organization, strong cooperative resource sharing, and nature-based (animistic) beliefs.
- Industrial Societies: Arising with the Industrial Revolution (late 18th to mid-20th century), these societies replaced manual and animal power with machine-based manufacturing and factories. This era was marked by rapid urbanization as rural populations migrated to cities for factory jobs, a capitalistic wage-labor economy, the polarization of social classes into the bourgeoisie

(capitalists) and the proletariat (workers), and major innovations like the steam engine, electricity, railways, and telegraphs.
- Post-Industrial Societies: Emerging in the late 20th century (1980s), these economies are driven by information, knowledge, technology, and services rather than physical manufacturing. Characteristics include high automation, computerization, robotics, artificial intelligence (AI), digital connectivity, globalization, and a highly educated workforce with greater emphasis on innovation and sustainability.

Question: Analyze how the Industrial Revolution fundamentally reshaped both social relations and natural systems. (5 Marks)

Answer:

The Industrial Revolution, starting in Britain during the mid-18th century, marked a massive structural shift from agricultural, cottage-based handcrafts to urban, machine-based factory production. While it catalyzed unprecedented economic growth, it severely disrupted both human social relations and natural ecosystems.

Reshaping of Social Relations:

- Rise of Social Classes: Industrialization split society into two competing classes: the wealthy bourgeoisie (factory owners and entrepreneurs) and a marginalized proletariat (working class), significantly widening wealth inequality and creating social tensions.
- Harsh Working Conditions: Workers, including women and children, endured extremely dangerous environments, low wages, and grueling 14–16 hour workdays, which eventually sparked labor unions, strikes, and movements like the Chartist Movement.
- Family Disruption: Shifting labor out of homes and farms and into factories disrupted traditional family units, as children and parents worked separate, long hours before child labor laws and reforms like the Factory Act of 1833 were introduced.

Destruction of Natural Systems:

- Air Pollution: The heavy combustion of fossil fuels (primarily coal) released massive amounts of CO_2, sulfur dioxide, and particulate matter, wrapping industrial hubs like London and Pittsburgh in toxic smog and causing acid rain.
- Water Pollution: Factories discharged untreated chemicals, dyes, sewage, and heavy metals into water bodies, decimating aquatic life and triggering deadly water-borne epidemics like cholera and typhoid (exemplified by London's "Great Stink" of 1858 along the Thames River).
- Land Degradation: Intensive mining of coal, iron, and minerals caused landscape fragmentation, toxic soil contamination, and massive deforestation to clear land for factories, mines, and expanding cities.

Question: Critically analyze the limitations of conventional GDP-based measures of development and explain what constitutes "Real Development." (5 Marks)

Answer:

Conventional development theory historically equated progress directly with economic growth, measuring success by rises in Gross Domestic Product (GDP), Gross National Product (GNP), and per capita income. While GDP is a highly useful indicator of market transactions and national productive capacity, it is an incredibly rough and flawed indicator of actual human well-being due to several fundamental limitations:

- Exclusion of Non-Market Activities: GDP ignores non-market contributions, such as unpaid household work and caregiving, which are vital to societal functioning.
- Masking of Inequality: National averages like GDP per capita hide severe wealth concentration. A country can experience surging economic growth while the vast majority of its population remains impoverished.
- Growth Without Livelihoods: Advancements in technology and automation can raise national income while simultaneously causing rising unemployment and job insecurity.
- Neglect of Environmental Degradation: Conventional GDP fails to deduct the immense environmental damage and depletion of natural capital caused by industrial growth, treating resource exploitation as profit rather than a cost.

In contrast, "Real Development" is a multidimensional, value-based process focused on human well-being. As Michael Todaro outlines, real development must involve deep structural, attitudinal, and institutional changes, the acceleration of economic growth, the reduction of inequality, and the absolute eradication of poverty. Its primary focus is people, not wealth. Real development is measured by improvements in life expectancy and health, access to quality education, community connection, individual freedom, security from violence and unemployment, and living in a clean, beautiful environment.

Question: Explain how current forms of economic development put strain on the natural environment through resource conversion, and identify the major environmental crises resulting from this process. (5 Marks)

Answer:

The current paradigm of economic development is highly linear and relies on converting raw natural resources (natural capital) into economic goods and infrastructure (cultural resources) to drive consumption and profit. Human survival and economic actions depend entirely on the environment. However, as societies expand their ecological footprints, they consume natural resources much faster than the Earth can replenish them, producing excessive waste and severe environmental crises.

The current form of uncontrolled economic development has triggered several interconnected global crises:

- Natural Resource Depletion: Rapid industrial consumption is depleting finite, non-renewable fossil fuels (coal, oil, gas) and minerals. Simultaneously, overconsumption and pollution are creating critical scarcities of renewable resources, particularly clean freshwater.

- Deforestation and Habitat Loss: Clearing vast forest lands for commercial logging, agriculture, and urbanization causes massive biodiversity loss, severe soil erosion, and disrupts regional hydrological cycles.
- Global Warming and Climate Change: The combustion of fossil fuels for energy and transport releases high concentrations of greenhouse gases (primarily CO_2, methane, and nitrous oxide). These gases trap solar heat, driving up global temperatures, shifting rainfall patterns, and causing rising sea levels that threaten vulnerable coastal nations.
- Soil and Water Pollution: Industrial dumping, untreated sewage, improper solid waste disposal, and heavy agricultural reliance on chemical pesticides and fertilizers have contaminated land and water bodies, drastically reducing soil fertility and usable water supplies.

Question: Define "Sustainable Development" according to the Brundtland Commission (1987) and analyze how its three core pillars relate to one another. (5 Marks)

Answer:

The concept of Sustainable Development was popularized by the UN's Brundtland Commission in its 1987 landmark report, Our Common Future. The commission defined it as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs". Rather than halting growth, it seeks to balance human aspirations with ecological limits, focusing on three key principles: human needs, appropriate technology, and intergenerational equity.

Sustainable development is structurally built on three highly integrated pillars:

- Economic Sustainability: This pillar encourages long-term economic growth, responsible resource management, fair trade, and investment in green innovation. It seeks to ensure prosperity while minimizing environmental degradation.
- Social Sustainability: This pillar promotes social equity, justice, human rights, and universal access to basic services like health and education. It strives to build cohesive, inclusive communities and improve the quality of life for all.
- Environmental Sustainability: This pillar focuses on protecting ecological integrity, preserving biodiversity, reducing pollution, and transitioning toward renewable energy sources.

The Interrelation of the Pillars:

These three pillars are deeply interdependent and cannot function in isolation. For instance, economic sustainability is impossible over the long term if the environmental pillar is neglected, as businesses will eventually collapse due to resource depletion. Similarly, social sustainability cannot be achieved without environmental protection, as the poor suffer most severely from environmental hazards, pollution, and climate change. True sustainable development is only realized at the intersection of all three pillars, ensuring economic actions are both socially inclusive and environmentally viable.

Question: Explain why the United Nations transitioned from the Millennium Development Goals (MDGs) to the Sustainable Development Goals (SDGs), and highlight the key characteristics of the SDGs. (5 Marks)

Answer:

In 2000, the UN established the Millennium Development Goals (MDGs), a set of 8 poverty-reduction targets with a deadline of 2015. While the MDGs catalyzed immense global progress—for example, helping Bangladesh drastically reduce extreme poverty and infant mortality while achieving gender parity in education—crucial development gaps persisted by 2015.

The reasons for the transition to the Sustainable Development Goals (SDGs) include:

- Persistent Inequality: Poverty, hunger, and wealth disparity remained highly severe both within and between nations.
- Ecological Crises: The MDGs did not adequately address environmental issues. Over time, climate change, rapid urbanization, and resource exploitation became urgent global crises requiring a dedicated, systemic response.
- Scope Exclusions: Crucial institutional aspects like peace, justice, governance, and sustainable consumption were omitted from the original MDG framework.

Consequently, in September 2015, the UN General Assembly adopted the 2030 Agenda, establishing the 17 SDGs supported by 169 targets. The major characteristics of the SDGs are:

- Universal Applicability: Unlike the MDGs, which focused almost exclusively on developing nations, the SDGs apply to all countries, rich and poor alike.
- Highly Integrated: They recognize that social development, economic growth, and environmental protection are deeply interconnected.
- Inclusivity: The framework is explicitly centered around the ethical promise to "leave no one behind" and actively reduce structural inequalities.
- The Five Ps Framework: The 17 goals are logically grouped around five key dimensions: People, Planet, Prosperity, Peace, and Partnership.

Question: Explain how renewable energy can mitigate global resource depletion and discuss the primary challenges faced during its implementation. (5 Marks)

Answer:

Renewable energy is energy harvested from natural, continuous processes that are naturally replenished on a human timescale, such as solar, wind, hydropower, biomass, geothermal, and ocean/tidal energy. Transitioning to renewable energy is vital to mitigating global resource depletion and climate change by:

- Preserving Finite Resources: It dramatically reduces humanity's dependence on rapidly depleting fossil fuels (coal, oil, natural gas).
- Slashing Carbon Emissions: Unlike fossil fuels, renewable sources generate little to no greenhouse gases during operation, reducing air pollution and global warming.

- Enhancing Energy Security: It diversifies national energy portfolios, lowering dependency on fuel imports and improving energy access in remote rural areas.

Primary Challenges in Renewable Energy Implementation:

- Intermittency: Solar and wind energy are highly variable and weather-dependent. This requires advanced, expensive energy storage systems (like large-scale batteries) to maintain grid stability.
- High Upfront Costs: While operational costs are extremely low, the initial capital investment required to build wind farms, solar fields, or geothermal facilities remains very high.
- Geographical Constraints: Renewable resources are unevenly distributed. Geothermal energy requires specific tectonic locations, wind turbines need open plains or coasts, and solar farms require vast sunlit land.
- Environmental Trade-offs: Renewable energy is not automatically sustainable. Large hydropower dams can alter aquatic ecosystems and displace local communities, while overharvesting local biomass can cause severe deforestation and soil erosion.

Question: Define the concept of "Ethical Consumption" and discuss the roles that individuals and institutions must play in achieving sustainable consumption. (5 Marks)

Answer:

Ethical consumption (or sustainable consumption) is an active consumer practice of purchasing goods and services that fulfill human needs while minimizing ecological harm, utilizing resources responsibly, and ensuring fair social and economic treatment of workers. It stands in direct opposition to modern hyper-consumerism, which has created severe global inequality (with the wealthiest 20% of the population consuming 17 times more energy and 145 times more cars than the poorest 20%) and put immense, unsustainable pressure on planetary resources. Ethical consumption requires looking beyond a product's price and quality to critically analyze how it was produced. A key example is mica mining in India; while mica is widely used to give cosmetics their sparkle, its extraction is heavily linked to dangerous child labor, making it an urgent ethical concern for global consumers.

Achieving sustainable consumption requires coordinated action across multiple levels:

- Individual Responsibility: Consumers must practice ethical decision-making before purchasing by asking: How was this produced? Were workers treated fairly? Can it be repaired or recycled? Individuals must practice the 3Rs (Reduce, Reuse, Recycle), purchase only what is necessary, buy durable products, and actively avoid single-use, high-waste goods.
- Institutional and Policy Responsibility: Businesses must enforce strict worker safety codes, ensure fair wages, adopt clean production methods, and provide transparent product supply-chain labeling. Governments must actively regulate harmful corporate practices, mandate environmental and social audits, restrict misleading consumer advertising, and invest in sustainable municipal waste and recycling programs.
- Social Responsibility: Fosters public dialogue to build a deeply rooted cultural norm of ethical, eco-friendly consumption and raise awareness of environmental and human impacts.

Question: Define "Sustainable Solid Waste Management". Discuss its key strategies and explain why implementing these practices is crucial for a rapidly urbanizing country like Bangladesh. (5 Marks)

Answer:

Sustainable Solid Waste Management refers to the systematic administration of activities associated with the generation, storage, collection, transfer, processing, and disposal of solid waste in a manner that is ecologically sound, economically viable, and socially acceptable. Rather than simply dumping waste in landfills, it treats waste as a potential resource to be recovered and reintegrated into the production cycle.

Key Strategies for Sustainable Waste Management:

- The 3Rs Hierarchy (Reduce, Reuse, Recycle): Minimizing waste generation at the source, finding secondary uses for existing products, and processing materials to create new raw products.
- Waste Segregation at Source: Separating biodegradable (organic) waste from non-biodegradable (plastic, glass, e-waste) materials at the household and commercial level to facilitate proper processing.
- Composting and Waste-to-Energy: Converting organic food waste into nutrient-rich compost for agriculture or bio-gas for clean energy generation.

Necessity in Bangladesh:

Rapid urbanization and high population density in cities like Dhaka and Chittagong generate massive volumes of domestic and commercial solid waste daily. Poorly managed waste accumulates in streets, clogs municipal drains (causing severe waterlogging during rain), pollutes surrounding rivers, and creates toxic dump sites that breed deadly diseases. Implementing sustainable waste practices is critical to preserving urban sanitation, protecting degrading water bodies like the Buriganga, reclaiming scarce land from sprawling landfills, and preventing toxic environmental pollution.

Question: Explain how Information and Communication Technology (ICT) can play a dual (positive and negative) role in Sustainable Development. As a CSE student, how would you design an initiative to address SDG 9 (Industry, Innovation, and Infrastructure)? (5 Marks)

Answer:

Information and Communication Technology (ICT) has a dual relationship with environmental and social sustainability:

- Negative Environmental Footprint: ICT lifecycles consume high amounts of electricity (especially data centers), increase carbon dioxide emissions, and generate massive amounts of toxic, non-biodegradable electronic waste (e-waste) as hardware is rapidly replaced.
- Positive Enabling Power: ICT acts as a powerful catalyst for green transitions by enabling smart farming, improving energy efficiency across power grids, optimizing waste collection logistics, and supporting big data and Geographic Information Systems (GIS) for real-time environmental monitoring.

Proposed CSE Initiative for SDG 9 (Resilient Infrastructure & Innovation):

To contribute to SDG 9, a CSE student can design an IoT-Enabled Smart Municipal Waste Management System.

- The Initiative: By integrating cheap, solar-powered ultrasonic sensors into city trash bins, the bins can communicate their real-time fill-levels to a centralized cloud platform.
- The Implementation: Using a routing algorithm, municipal waste collection trucks can dynamically navigate only to those bins that are 90% or more full.
- The Sustainable Outcome: This directly reduces fuel consumption, slashes greenhouse gas emissions from municipal fleets, prevents overflowing bins from causing public health hazards, and utilizes data analytics to plan future urban waste infrastructure efficiently.

Question: Differentiate between Normative Relativism and Universalism in ethics. Explain how an engineer can use Utilitarianism to resolve an ethical dilemma when executing a public infrastructure project. (5 Marks)

Answer:

In engineering ethics, professionals must navigate conflicting values using established ethical theories.

- Normative Relativism: This theory argues that moral values and norms are relative to specific cultures, societies, situations, and times. It claims there are no universal, absolute moral guidelines independent of social context. For example, a safety standard accepted in one country might be considered lax or unacceptable in another.
- Universalism: In contrast, universalism asserts that certain fundamental moral norms and values (such as the duty to protect human life and prevent avoidable harm) are universally applicable to all people, regardless of their cultural background, location, or time.

Application of Utilitarianism to an Engineering Dilemma:

Utilitarianism evaluates the morality of an action based on its overall consequences. Guided by the Greatest Happiness Principle, the preferred ethical action is the one that maximizes positive outcomes (benefits, safety, happiness) for the largest number of people while minimizing harm and suffering.

- The Dilemma: An engineer is designing a major highway that will bypass a highly congested city center, saving transit time and reducing air pollution for millions, but its construction will displace a small farming village and cause localized environmental damage.
- The Resolution: A utilitarian engineer would calculate the net positive outcomes. Since the project benefits millions by reducing emissions and traffic accidents, the project is deemed morally permissible. However, to minimize the negative consequences (as required by the principle), the engineer must ensure the displaced villagers are fairly compensated with modern housing, and implement green corridors to mitigate local ecological damage.